

Slide 1 – Title

Road Safety & Emergency Accessibility Prediction System

Final Year Data Science Project Proposal

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Slide 2 – Introduction

- Road safety and emergency accessibility are important concerns during disasters and emergencies.
 - Existing navigation systems mainly focus on shortest or fastest routes.
 - They usually do not consider:
 - Road conditions
 - Flood-prone areas
 - Emergency accessibility
 - Infrastructure risks
 - This project introduces a data-driven system that predicts road safety and analyzes emergency accessibility using geospatial datasets.
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Slide 3 – Current Situation Leading to Problem Identification

Problems Identified:

- Some roads become unsafe during floods and bad weather.
- Poor road conditions increase travel risks.
- Emergency vehicles may experience delays due to inaccessible roads.
- Some regions have limited hospital accessibility.
- Existing systems lack safety-aware and emergency-focused analysis.

Existing System Limitations:

- Mainly focus on distance and travel time.

- Limited road risk and accessibility analysis.
 - No integrated road safety prediction system.
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Slide 4 – Proposed Technique to Solve the Current Problem

Proposed Solution:

Develop a multimodal geospatial data science system that:

- Predicts road safety levels
- Analyzes emergency accessibility
- Recommends safer travel routes
- Identifies high-risk road segments

Techniques Used:

- Data preprocessing and feature engineering
- Geospatial analytics
- Machine learning prediction
- Recommendation system
- Interactive GIS map visualization

Machine Learning Models:

- Logistic Regression
 - Random Forest
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Slide 5 – Feasibility

Technical Feasibility:

- Open-source datasets are available
- Python libraries support geospatial analysis
- Can run on a low-end computer

Economic Feasibility:

- Fully based on free tools and datasets
- No paid APIs or cloud services required

Operational Feasibility:

- Useful for emergency accessibility and travel safety analysis
 - Practical real-world application
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Slide 6 – Project Description

This project is a web-based geospatial analysis and prediction system that uses:

- Road datasets
- Healthcare accessibility datasets
- Flood and vulnerability indicators
- Points of interest datasets

to analyze:

- Road safety conditions
- Emergency healthcare accessibility
- Safer travel recommendations

The system visualizes results using interactive maps and dashboards.

Slide 7 – Deliverables**Final Deliverables:**

- Road safety prediction model
- Emergency accessibility analysis system
- Safe route recommendation module
- Interactive GIS map dashboard
- Final research report and documentation

Analysis Outputs:

- Safe vs risky road identification
 - Hospital accessibility analysis
 - Flood-prone route analysis
 - Safer travel recommendations
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Slide 8 – Resources Required

Software & Tools:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- GeoPandas
- Folium
- Scikit-learn
- Flask
- SQLite

Datasets:

- OpenStreetMap Roads Dataset
- Points of Interest Dataset
- Healthcare Accessibility Dataset
- World Bank Development Dataset

Hardware:

- Standard laptop or desktop
- Internet connection

Slide 9 – Expected Output and Outcome

Expected Outputs:

- Road safety prediction results
- Emergency accessibility analysis
- Safer route recommendations
- GIS-based visualizations and dashboards

Expected Outcomes:

- Improved awareness of road and infrastructure risks
- Better emergency accessibility understanding
- Data-driven travel safety recommendations
- Support for disaster preparedness and planning

Slide 10 – Time Plan for Implementation

Gantt Chart

(Insert Gantt Chart Here)

Slide 11 – Limitations

- No real-time traffic data integration
- Accuracy depends on dataset quality
- Some rural road data may be incomplete
- Flood conditions may change dynamically
- Limited by available infrastructure datasets
- Simplified route recommendation compared to commercial navigation systems